

A machine learning approach to detecting anomalies in MWA observations

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ABSTRACT

Detecting the cosmological 21 cm signal from the Epoch of Reionization (EoR) remains one of the most challenging objectives in modern radio astronomy. The signal is extremely faint and highly susceptible to contamination arising from instrumental artefacts, calibration failures, radio frequency interference, and other observational systematics. Undetected anomalies within observational datasets can bias downstream scientific analysis and potentially lead to false astrophysical interpretations.

This study investigates the use of machine learning approaches for identifying systematically contaminated observations using calibration-derived metrics obtained during radio telescope processing pipelines. Both unsupervised anomaly detection and supervised classification techniques were evaluated using a dataset containing 12,298 observations characterized by calibration quality metrics. Initial experiments using Isolation Forest demonstrated limited sensitivity in detecting subtle anomalies due to class imbalance and overlapping feature distributions. Supervised approaches including XGBoost, LightGBM, logistic regression, and random forest achieved substantially improved detection performance.

To improve sensitivity toward hidden systematics not captured by existing labels, a hybrid anomaly detection framework combining Isolation Forest, Local Outlier Factor, and Principal Component Analysis reconstruction error is proposed. Results demonstrate that calibration metrics contain meaningful information for identifying systematically contaminated observations and highlight the importance of automated anomaly detection frameworks for improving the reliability of future EoR observations.

1. INTRODUCTION

One of the primary goals of contemporary radio astronomy is the detection of the redshifted 21 cm hydrogen signal originating from the Epoch of Reionization, a period marking the formation of the first luminous structures in the early universe. Experiments associated with instruments such as the Square Kilometre Array Observatory (SKA), Murchison Widefield Array (MWA), and other low-frequency interferometers aim to measure this signal.

However, the EoR signal is several orders of magnitude weaker than astrophysical foregrounds and highly vulnerable to contamination introduced by systematic effects. Calibration errors, unstable instrumental gain solutions, receiver noise variation, and incomplete convergence during signal processing can introduce biases that propagate into later stages of analysis.

Traditional quality control procedures often rely on manual inspection or predefined threshold-based filtering. These approaches become increasingly inefficient as observational datasets continue to grow in scale. Machine learning provides a potential solution by enabling automated identification of anomalous observations that may contain hidden systematic contamination.

This work explores the use of machine learning models to detect systematically dominated observations using calibration-derived quality metrics extracted from radio astronomy processing pipelines.

2. MOTIVATION

Unlike conventional radio astronomy imaging, EoR experiments rely on statistical detection of an extremely weak signal buried beneath bright foreground emission. The scientific challenge is not simply removing foregrounds, but ensuring that instrumental systematics do not create artificial spectral signatures that become indistinguishable from the cosmological signal.

Calibration anomalies can affect EoR science in several ways:

Spectral contamination from gain errors introduces frequency-dependent artifacts that leak into high-delay Fourier modes. Incomplete calibration convergence results in incorrect antenna gain solutions that propagate systematic errors into visibilities. Receiver instabilities introduce time-dependent fluctuations that bias power spectrum estimation. Baseline corruption affects uv-coverage consistency and distorts image reconstruction. Residual systematic structures can mimic large-scale 21 cm fluctuations, producing false positive detections.

Since EoR power spectrum estimation assumes extremely well-characterized instrument behaviour, even a small number of poor-quality observations can bias final cosmological measurements.

Traditional quality control methods often rely on manual inspection or hard-threshold rejection criteria. However, modern telescopes generate observational datasets at scales where manual analysis becomes impractical. Automated anomaly detection therefore becomes essential.

3. DATASET AND CALIBRATION METRICS

This study used 12,298 observations obtained during calibration processing stages. Several calibration diagnostics were extracted as machine learning features representing instrumental and calibration stability.

The selected features include:

- *RMSCONVG*: Root mean square convergence metric measuring calibration stability.
- *SKEW*: Statistical skewness of calibration solutions, indicating asymmetry or instability in gain distributions.
- *RECVVAR*: Receiver variance across antennas and frequency channels representing instrumental noise fluctuations.
- *DFFTPOW*: Delay-domain Fourier transform power sensitive to spectral structure contamination.
- *BLUNUSEDFRAC*: Fraction of interferometric baselines rejected during calibration.
- *NONCONVGCHSFRAC*: Fraction of frequency channels where calibration solutions failed to converge.

These metrics provide a quantitative description of calibration quality and potential systematic contamination. The dataset is inherently imbalanced, consisting of:

- Good observations: 11,108 (90.3%)
- Failed observations: 1,190 (9.7%)

This class imbalance presents a significant challenge for anomaly detection.

4. UNSUPERVISED ANOMALY DETECTION USING ISOLATION FOREST

The first stage of this study employed the Isolation Forest algorithm, an unsupervised anomaly detection technique specifically designed to identify rare and abnormal observations without requiring labeled training data. Unlike conventional classification algorithms, Isolation Forest operates on the principle that anomalous observations are “few and different”. Rather than learning a decision boundary between classes, the algorithm isolates observations by recursively partitioning the feature space using randomly selected features and split values.

The algorithm constructs an ensemble of binary decision trees, referred to as isolation trees. During tree construction, each split randomly selects:

- A feature from the available feature set
- A random split value within the range of that feature

An observation is considered anomalous if it can be isolated after a relatively small number of splits. In contrast, normal observations located within dense clusters require a greater number of partitioning steps. The anomaly score is determined from the average path length across all isolation trees. Mathematically, the anomaly score is defined as

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}} \quad (1)$$

Table 1. Isolation Forest Confusion Matrix

	Predicted Normal	Predicted Anomaly
Actual Normal	10384	878
Actual Anomaly	684	366

Table 2. Isolation Forest Performance Metrics

	1.00	1.00	1.00
Random Forest	1.00	0.00	0.00
Logistic Regression	1.00	0.96	0.98
XGBoost	1.00	1.00	1.00
LightGBM	0.03	0.43	0.05
Isolation Forest			

where $h(x)$ is the path length required to isolate observation x , $E(h(x))$ is the average path length over all trees and $c(n)$ is the average path length normalization factor for a dataset of size n . Observations with shorter path lengths produce anomaly scores closer to 1 and are therefore classified as anomalous.

Isolation Forest was selected because calibration failures in radio astronomy often manifest as unusual combinations of calibration metrics that deviate from the expected statistical behavior of normal observations. The algorithm is computationally efficient and suitable for large-scale observational datasets where labeled anomaly examples may be limited.

Initial contamination estimation was set at 5.5%. Performance evaluation yielded the confusion matrix illustrated in Table 4. The resulting performance metrics were: Precision of 46%, Recall of 26% and F1-Score of 33%.

Several limitations were observed during unsupervised anomaly detection. First, the dataset exhibits severe class imbalance, causing anomalous observations to remain underrepresented during tree construction. Second, many systematic failures appear to be subtle rather than extreme outliers, reducing the effectiveness of purely distance-based anomaly detection.

Increasing contamination estimation to the true anomaly rate of 9.7% improved recall to 42%, but introduced additional false positives. Application of robust scaling produced minimal improvement. These results suggest that calibration failures are not always represented as obvious outliers in feature space.

4.1. Supervised Classification

Although unsupervised anomaly detection can identify unusual observations, many systematic failures in radio astronomy calibration pipelines are subtle and may not appear as extreme statistical outliers. To improve detection performance, supervised classification algorithms were trained using labeled observations categorized as PASS (scientifically acceptable) and FAIL (contaminated observations).

The objective of supervised learning is to learn the relationship between calibration metrics and known observational quality outcomes, allowing the model to predict whether previously unseen observations are likely to contain systematic contamination. The dataset was partitioned using an 80:20 training-testing split. Several classification algorithms were evaluated:

1. Random Forest

2. Logistic Regression

3. XGBoost

4. LightGBM

5. Isolation Forest (baseline comparison)

Performance results are summarized in Table 4.1.

4.2. Random Forest

Random Forest is an ensemble learning algorithm that constructs multiple decision trees during training and combines their predictions through majority voting. Each decision tree is trained using a bootstrap sample of the original dataset, while feature selection at each split is randomized. This process reduces overfitting and improves model generalization. For classification, the final prediction is obtained by

$$\hat{y} = \text{mode}(T_1(x), T_2(x), \dots, T_n(x)) \quad (2)$$

where $T_i(x)$ represents the prediction from the i^{th} decision tree.

Random Forest is particularly suitable for this problem because it can capture complex nonlinear relationships between calibration metrics while remaining relatively robust to noisy data and feature interactions. Tree-based ensemble methods significantly outperformed unsupervised detection approaches. However, the near-perfect classification performance raises the possibility of data leakage, feature redundancy, or overly correlated labels and requires further validation.

4.3. Logistic Regression

Logistic Regression is a probabilistic linear classification algorithm used to estimate the likelihood of an observation belonging to a given class. Unlike tree-based methods, Logistic Regression assumes a linear relationship between input features and the log-odds of class membership. The probability of class membership is modeled using the sigmoid function

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}} \quad (3)$$

where β_i are learned coefficients corresponding to each feature.

This model serves as a baseline classifier to evaluate whether systematic contamination can be linearly separated using calibration metrics. The poor performance observed suggests that relationships between calibration metrics and failure states are highly nonlinear.

4.4. XGBoost

Extreme Gradient Boosting (XGBoost) is an advanced ensemble learning algorithm based on gradient boosting decision trees. Unlike Random Forest, where trees are constructed independently, XGBoost builds trees sequentially. Each new tree attempts to correct errors made by previous trees by minimizing a differentiable loss function.

At each iteration, the model updates according to

$$F_m(x) = F_{m-1}(x) + \eta h_m(x) \quad (4)$$

where $F_m(x)$ is the updated model, $h_m(x)$ is the newly trained decision tree and η is the learning rate controlling update magnitude.

The optimization objective includes both prediction loss and regularization

$$L = \sum l(y_i, \hat{y}_i) + \sum \Omega(f_k) \quad (5)$$

where Ω penalizes model complexity.

XGBoost is particularly effective for identifying systematic contamination because it captures complex nonlinear relationships while handling noisy, heterogeneous, and imbalanced datasets effectively.

4.5. LightGBM

Light Gradient Boosting Machine (LightGBM) is a high-performance gradient boosting framework optimized for computational efficiency. Similar to XGBoost, LightGBM builds decision trees sequentially to minimize prediction error. However, it differs in two important ways. First, LightGBM uses histogram-based feature binning to reduce memory usage and computational cost.

Second, instead of growing trees level-wise, LightGBM grows trees leaf-wise, selecting the leaf that maximally reduces loss. The algorithm optimizes an objective function of the form

$$L = \sum l(y_i, \hat{y}_i) + \lambda \Omega(T) \quad (6)$$

where $\Omega(T)$ represents tree complexity regularization.

LightGBM is particularly useful for large astronomical datasets because radio telescope observations can generate millions of samples, requiring efficient model training while maintaining predictive accuracy.

5. HYBRID SYSTEMATICS DETECTION FRAMEWORK

Although supervised learning performs well on known failure patterns, future observations may contain previously unseen systematic contamination. To address this limitation, a hybrid anomaly detection framework was designed using only trusted observations where $PASS = \text{True}$. The framework models normal system behaviour and identifies deviations from learned patterns. Three complementary anomaly detectors were combined: Isolation Forest, Local Outlier Factor (LOF) and Principal Component Analysis Reconstruction Error.

5.1. Isolation Forest Anomaly Scoring

Isolation Forest constructs an ensemble of randomly generated binary trees that recursively partition the feature space. The key idea is that anomalies are *easier to isolate* than normal observations. For a given observation x , the algorithm computes the path length $h(x)$, defined as the number of splits required to isolate the point within a tree. The expected path length over an ensemble of trees is given by:

$$E[h(x)] = \frac{1}{T} \sum_{t=1}^T h_t(x), \quad (7)$$

where T is the number of trees. The anomaly score is then defined as:

$$S_{\text{IF}}(x) = 2^{-\frac{E[h(x)]}{c(n)}}, \quad (8)$$

where $c(n)$ is a normalization constant representing the average path length of unsuccessful searches in a binary search tree:

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n}, \quad (9)$$

and $H(i)$ is the i -th harmonic number. Values of $S_{\text{IF}}(x)$ close to 1 indicate strong anomalies, while values near 0.5 correspond to typical observations within dense regions of the data distribution.

The trained anomaly detection framework was applied to the full observational dataset. For each observation, the following anomaly scores were computed:

- Isolation Forest anomaly score
- LOF score
- PCA reconstruction error

5.2. Local Outlier Factor (LOF)

Local Outlier Factor is a density-based anomaly detection algorithm that identifies observations whose local neighborhood density differs significantly from surrounding observations. Unlike Isolation Forest, which identifies global anomalies, LOF focuses on local inconsistencies. For a given observation, the local density is estimated based on distances to its nearest neighbors. The anomaly score is computed as

$$LOF(x) = \frac{\sum_{o \in N(x)} \frac{lrd(o)}{lrd(x)}}{|N(x)|} \quad (10)$$

where $N(x)$ represents neighboring observations and $lrd(x)$ is local reachability density. Large LOF values indicate abnormal local structure. LOF is useful because subtle calibration failures may not appear globally anomalous but may differ significantly from nearby observations.

5.3. Principal Component Analysis Reconstruction Error

Principal Component Analysis (PCA) is used to learn the underlying geometric structure of normal observations. PCA reduces the dimensionality of the feature space by projecting observations onto orthogonal principal components that capture maximum variance. After learning the reduced representation, each observation is reconstructed back to the original feature space. The reconstruction error is computed as

$$Error = ||x - \hat{x}||^2 \quad (11)$$

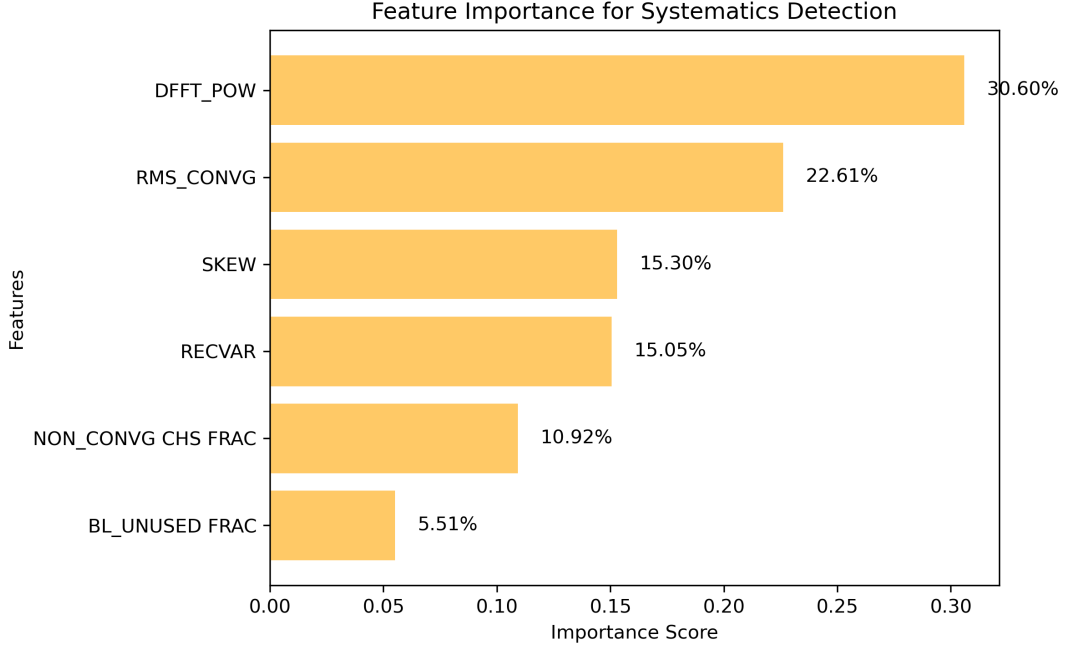


Figure 1. Contribution of the different calibration parameters in reaching the outcomes.

where x is the original observation and $\hat{x}$ is the reconstructed observation.

Observations with large reconstruction errors deviate strongly from the learned structure and may indicate hidden systematic contamination. In radio astronomy, PCA reconstruction error is particularly useful because subtle systematic failures may disrupt the normal geometric relationships between calibration metrics. Measures structural deviation from the learned feature geometry.

5.4. Combined Systematics Score

Each anomaly detector captures different aspects of abnormal behavior. To leverage the complementary strengths of each method, anomaly scores were combined into a single systematics score. The final score was defined as

$$\text{SystematicsScore} = 0.4(\text{IsolationForest}) + 0.3(\text{LOF}) + 0.3(\text{PCAError}). \quad (12)$$

Observations with scores exceeding the 95th percentile were flagged for further scientific inspection.

6. FEATURE IMPORTANCE ANALYSIS

Feature importance analysis was performed to quantify the relative contribution of each calibration metric toward anomaly detection. Features with higher importance indicate stronger predictive capability in distinguishing normal observations from systematically contaminated observations. The analysis demonstrates that calibration-derived metrics carry substantial information regarding observational quality and can serve as effective indicators of systematic contamination.

7. SCIENTIFIC IMPLICATIONS FOR EPOCH OF REIONIZATION STUDIES

In EoR science, even small systematic contamination can significantly bias power spectrum estimation and interfere with extraction of the faint cosmological 21 cm signal. Undetected systematic errors may:

- Introduce false astrophysical signatures
- Suppress weak cosmological signals
- Bias foreground subtraction procedures

- Propagate uncertainty into downstream scientific interpretation

Automated anomaly detection frameworks provide an essential safeguard by identifying problematic observations before inclusion in final scientific analysis. Such approaches will become increasingly important for next-generation instruments such as the Square Kilometre Array Observatory.

8. CONCLUSION

This study demonstrates the feasibility of applying machine learning techniques to detect systematic contamination in radio astronomy calibration datasets. Results show that:

- Calibration metrics provide meaningful information regarding observational quality.
- Unsupervised anomaly detection alone struggles with subtle systematic failures.
- Supervised tree-based classifiers achieve significantly improved performance.
- Hybrid anomaly detection frameworks can identify hidden systematic contamination beyond existing labels.

Future work should focus on validating the framework using Phase III observations and extending the anomaly detection system to incorporate additional observational and instrumental metrics. Reliable identification of systematically contaminated observations represents a critical step toward improving the robustness of EoR signal detection in future radio astronomy experiments.

REFERENCES